PHYS 161 Notes

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1 Units, Physical Quantities, and Vectors

1.1 SI Units

- **SI Units** refer to system of units used by engineers and scientists across the world.
- For PHYS 161, these will be the most commonly used units in problems.

Application	Unit	Shorthand
Time	Seconds	s
Length	Meter	m
Mass	Kilogram	kg

- Additionally, many prefixes may be used in problems, necessitating conversions.

Length	Mass	Time
1 nanometer $= 1$ nm $= 10^{-9}$ m	1 microgram $= 1\ \mu$g $= 10^{-9}$ kg	1 nanosecond$= 1$ ns $= 10^{-9}$ s
1 micrometer $= 1\ \mu$m $= 10^{-6}$ m	1 milligram $= 1$ mg $= 10^{-6}$ kg	1 microsecond $= 1\ \mu$s $= 10^{-6}$ s
1 millimeter $= 1$ mm $= 10^{-3}$ m	1 gram $= 1$ g $= 10^{-3}$ kg	1 millisecond $= 1$ ms $= 10^{-3}$ s

1.2 Unit Conversions

- In equations, quantities may only be added if they have the same dimension.
- When multiplying two quantities, the product's dimension is the product of the operands' dimensions.

$$\left(\frac{\mathrm{m}}{\mathrm{s}}\right)(\mathrm{s}) = \mathrm{m}$$

1.3 Significant Figures

- The **uncertainty** of a measurement is often indicated by the number of important digits, known as the **significant figures.**
  - In provided values, all non-zero digits are significant figures.

$$525 \Rightarrow \text{three significant figures}$$

  - Meanwhile, zero digits are only significant if they are between non-zero digits, or they are trailing non-zero digits when there is a decimal point.

$$40001 \Rightarrow \text{five significant figures}$$
$$120.0 \Rightarrow \text{four significant figures}$$

- When adding quantities, the significant figures of the sum is equal to the operand with the least significant figures.

$$25.7 + 13 = 38.7 \approx 39 \Rightarrow \text{two significant figures}$$

- When multiplying quantities, the same rule applies to the product.

$$(13.7)(20.95) = 287.015 \approx 287 \Rightarrow \text{three significant figures}$$

- In longer physics problems with numerous equations, using the actual values in calculations and only converting the final answer to the correct amount of significant figures is best practice.

2 Motion Along a Straight Line

2.1 Displacement, Time, Velocity, and Acceleration

- **Time** is measured in seconds, and usually refers to the time passed between two particular points in time.
  - Denoted t.
- **Displacement** and **distance** is measured in meters.
  - Denoted Δx and x, respectively, along the x-axis.
- **Velocity** is measured in meters per second.
  - Denoted v.
- **Acceleration** is measured in meters per square second.
  - Denoted a.

$$\Delta x = x_2 - x_1$$
$$v_{\text{av}} = \frac{\Delta x}{\Delta t}$$
$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}$$
$$a_{\text{av}} = \frac{\Delta v}{\Delta t}$$
$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{\mathrm{d}v}{\mathrm{d}t}$$

2.2 Motion with Constant Acceleration

- With constant acceleration, the following **kinematic equations** hold true:

$$v = v_0 + a\Delta t$$
$$\Delta x = v_0 \Delta t + \frac{1}{2} a \Delta t^2$$
$$v^2 = v_0^2 + 2a\Delta x$$

Derivation: Displacement Kinematic Equation

$$v = v_0 + a\Delta t$$
$$\int_{t_1}^{t_2} v \,\mathrm{d}t = \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t$$
$$\int_{t_1}^{t_2} \frac{\mathrm{d}x}{\mathrm{d}t} \,\mathrm{d}t = \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t$$
$$\int_{x_1}^{x_2} \mathrm{d}x = \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t$$
$$\Delta x = v_0 \Delta t + \frac{1}{2} a \Delta t^2$$

3 Motion in Two or Three Dimensions

3.1 Vector Quantities

- Many of the definitions from one-dimensional motion hold true in two or three dimensions, albeit using vectors.

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$
$$\vec{v} = v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k}$$
$$\vec{a} = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}$$

3.2 Projectile Motion

- In **projectile motion**, an object in the air only experiences acceleration due to gravity.
- Because the same kinematic equations can be applied *per component* of the vectors, the following generalizations can be made (given that z is the vertical direction):

$$v_x = v_{0x}$$
$$\Delta x = v_{0x}\Delta t$$

$$v_y = v_{0y}$$
$$\Delta y = v_{0y}\Delta t$$

$$v_z = v_{0z} - g\Delta t$$
$$\Delta z = v_{0z}\Delta t - \frac{1}{2}g\Delta t^2$$
$$v_z^2 = v_{0z}^2 - 2g\Delta z$$

Constant g is the acceleration due to Earth's gravity, $9.81\ \mathrm{m/s^2}$. Note that y is generally the vertical direction when dealing with two dimensions.

- Given these equations, we can also derive a general formula for a projectile's **range.**

Derivation: Range Formula

Let Δx equal R and let Δy equal 0.

$$v_y = v_{0y} - gt$$
$$gt = v_{0y} - v_y$$
$$t = \frac{v_0 \sin\theta - v\sin\theta}{g}$$
$$t = \frac{v_0 \sin\theta - (-v_0 \sin\theta)}{g}$$
$$t = \frac{2v_0 \sin\theta}{g}$$

By substitution

$$R = v_{0x}t$$
$$R = v_0 \cos\theta t$$
$$R = v_0 \cos\theta \frac{2v_0 \sin\theta}{g}$$
$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

3.3 Circular Motion

- Even if a particle's speed is constant, its velocity cannot be constant in circular motion because *its direction is constantly changing.*
- The direction of the velocity changes due to acceleration directed toward the center of its rotation, known as **centripetal acceleration.**
  - Denoted a_t or a_{rad}.
  - Thus, when centripetal acceleration is the only acceleration vector acting on a particle, the particle undergoes **uniform circular motion** because it will always be perpendicular to the particle's velocity vector.
- An acceleration vector parallel to the particle's velocity vector is known as **tangential acceleration.**
  - Denoted a_t.
  - This is known as **nonuniform circular motion**, because the speed of the particle will also be changing.

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t}$$

- **Frequency** is the number of revolutions (full rotations) per second.
  - Denoted f.
- **Period** is the number of seconds needed to make one revolution.
  - Denoted T.

$$T = \frac{1}{f}$$

- Given radius of the circular motion R

$$v = \frac{2\pi R}{T}$$
$$a_c = \frac{v^2}{R}$$

3.4 Relative Motion

- Given reference frames A, B, and C where AB means "A in the reference frame of B"

$$\vec{v}_{AB} + \vec{v}_{BC} = \vec{v}_{AC}$$
$$\vec{v}_{AB} = -\vec{v}_{BA}$$

- The **Law of Sines** and the **Law of Cosines** are useful formulas for relative motion problems whenever angles are needed.

Definition: Law of Sines

Given sides a, b, and c opposite of angles A, B, and C, respectively,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Definition: Law of Cosines

Given sides a, b, and c opposite of angles A, B, and C, respectively,

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

4 Newton's Laws of Motion

4.1 Newton's First Law

Definition: Newton's First Law

An object subject to no external forces is at rest or moves with constant velocity if viewed from an inertial reference frame.

$$\text{Inertial Reference Frame} \equiv \text{Reference frame in which Newton's laws are valid}$$

- An inertial reference frame must not be accelerating or rotating.

4.2 Newton's Second Law

- **Forces** are pushes or pulls on objects, measured in Newtons (N) or $(\mathrm{kg} \cdot \mathrm{m/s}^2)$.

Definition: Newton's Second Law

$$\Sigma \vec{F} = m\vec{a}$$

4.3 Newton's Third Law

Definition: Newton's Third Law

For every action there is an equal and opposite reaction.

Given objects A and B where AB means "exerted from A on B"

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

- For example, when two objects collide, they experience the same magnitude of force from the collision.

5 Applying Newton's Laws

5.1 Types of Forces

- **Fundamental forces** are the four natural interactions that govern how particles interact.
  - **Weight**
  - Electromagnetic
  - Weak nuclear
  - Strong nuclear
- **Non-fundamental forces** are forces whose interactions are derived from the fundamental forces.
  - **Spring**
  - **Contact** (tension, normal, friction, ...)
  - Microscopic
- The **weight** force is from gravity. On Earth, it is defined as follows:

$$\vec{w} = m\vec{g}$$

- However, there are also exists a general definition for weight force, the **Universal Law of Gravitation** using the gravitational constant G

$$G = 6.6743 \times 10^{-11}\ \mathrm{m}^3 \cdot \mathrm{kg}^{-1} \cdot \mathrm{s}^{-1}$$

Definition: Universal Law of Gravitation

The force between any two objects is defined as follows (given that r is the distance between the objects' centers):

$$F_1 = F_2 = G\frac{m_1 m_2}{r^2}$$

- The **spring force** is from compressing a spring.
  - Springs have differing **spring constants** usually denoted as k, and they define the ratio between the force exerted by the spring and the distance compressed from the equilibrium position.

$$\vec{F}_{\text{spr}} = -k\vec{x}$$

- The **normal force** is exerted by a surface on an object.
  - Generally denoted $\vec{n}$.
- The **tension force** is the pulling force exerted equally throughout a rope, cable, etc.
  - Generally denoted $\vec{T}$.
- The **friction force** is exerted by a surface on an object and acts perpendicular to the normal force. It depends on the roughness of the surface (expressed by the coefficient of friction).
  - **Kinetic friction**, denoted $\vec{f}_k$ with coefficient of kinetic friction μ_k, acts on objects sliding across a surface.

$$\vec{f}_k = \mu_k \vec{n}$$

  - **Static friction**, denoted $\vec{f}_s$ with coefficient of static friction μ_s, acts on still objects until the applied force exceeds the maximum static friction.

$$\vec{f}_s \leq \vec{f}_{s,\max} = \mu_s \vec{n}$$

5.2 Forces in Equilibrium

- Objects are in equilibrium when they are at rest or moving with constant velocity, meaning that the net force acting on those objects must be zero.

$$\begin{aligned} \Sigma \vec{F} &= 0 \\ &\Rightarrow \\ \Sigma F_x &= 0 \\ \Sigma F_y &= 0 \end{aligned}$$

- Thus, unknown forces can be solved in terms of other forces very easily.

5.3 Dynamics of Circular Motion

- Based on Newton's Second Law, circular motion can result from many different forces.

$$\Sigma F = ma_c = \frac{mv^2}{R}$$

- As a result, denoting a singular "centrifugal force" is generally not accurate.
  - For example, if an object is moving vertically with uniform circular motion, the weight force points toward the center of rotation at the top, but away from the center of rotation at the bottom.
- Additionally, direction should be considered relative to the center of rotation.

6 Work and Kinetic Energy

6.1 Work

- A force does **work** on a body if the body undergoes a displacement.
- The units of work are joules (J) or newton-meters.

Definition: Work

Given work W, force $\vec{F}$, and displacement $\vec{s}$

$$W = \vec{F} \cdot \vec{s} = Fs\cos(\theta)$$

- Thus, it follows that a force does **negative work** when applied opposite of the direction of $\vec{s}$ and does **zero work** when applied perpendicular to the direction of $\vec{s}$.
- Additionally, given this definition, there are multiple ways to define **net work.**

$$W_{\text{net}} = \Sigma\vec{F} \cdot \vec{s} = \Sigma W$$

6.2 Kinetic Energy

- **Energy** is the capacity to do work, and also uses joules.
- **Kinetic energy** is the energy of a body from its mass and speed.

Definition: Kinetic Energy

Given kinetic energy K, mass m, and speed v

$$K = \frac{1}{2}mv^2$$

- Because kinetic energy is based off of speed, it is either positive or zero.
- Additionally, work may be expressed as the **change in kinetic energy.**

$$W = \Delta K$$

Derivation: Work-Energy Theorem

Given work W and kinetic energy K

$$F_{\parallel} = ma = m\frac{v_2^2 - v_1^2}{2s}$$

$$F_{\parallel}s = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

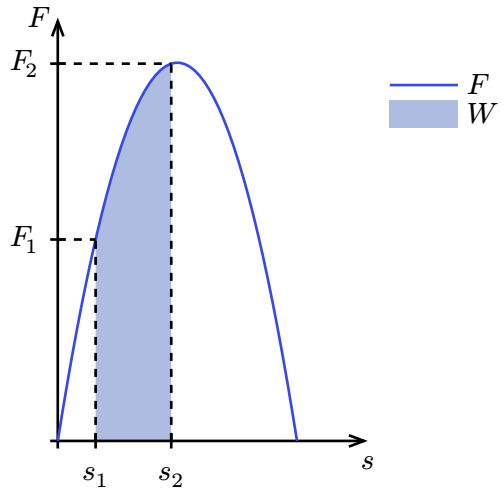
$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \Delta K$$

- However, these formulas only hold true when the force remains constant during the displacement.

6.3 Work and Energy with Varying Force

- If an object is experiencing a varying net force, then the total work can be approximated using the average force per increment of the total displacement.
- Thus, as the increment tends to zero, work may be defined as follows:

$$W = \int_{s_1}^{s_2} F \, \mathrm{d}s$$



6.4 Power

- **Power** is the rate that work is done, measured in watts (W) or joules per second (J/s).
  - Denoted P.
- Thus, average power is defined as follows:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t}$$

Definition: Power

Given power P, work W, velocity $\vec{v}$, and force $\vec{F}$

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \vec{F} \cdot \vec{v} = Fv\cos\theta$$

7 Potential Energy and Energy Conservation

- **Potential energy** is energy based on position, and may be *converted* into other forms of energy depending on the change in position.
  - For example, a fruit hanging from a tall tree has the *potential* to reach a higher speed (and therefore kinetic energy) before it hits the ground than a fruit hanging from a smaller tree.

7.1 Gravitational Potential Energy

- **Gravitational potential energy** is potential energy dependent on an object's mass and height about the ground.
  - Denoted U_g.

Definition: Gravitational Potential Energy

Given gravitational potential energy U_g, object's mass m, acceleration due to gravity g, and height above the ground h

$$U_g = mgh$$

- Notice how an object's gravitational potential energy *decreases* as its height above the ground decreases even though the work done by the weight force on a falling object is *positive*.
- Thus, the work done by gravity is equal to the negative change in gravitational potential energy.

$$W_g = -\Delta U_g$$

7.2 Elastic Potential Energy

- **Elastic potential energy** is potential energy stored in deformable objects such as springs when they are stretched or compressed from equilibrium.
  - Denoted U_s.

Definition: Elastic Potential Energy

Given elastic potential energy U_s, spring constant k, and elongation of object x

$$U_s = \frac{1}{2}kx^2$$

- Similarly to work done by gravity, because an object's elastic potential energy increases the more it is displaced from equilibrium, the work done by the spring force is equal to the negative change in elastic potential energy.

$$W_s = -\Delta U_s$$

7.3 Conservative and Nonconservative forces

- **Mechanical energy** is energy that is either kinetic or potential, and the total mechanical energy of a system is only constant
- **Conservative forces** do work independent of the path, meaning that they only depend on the endpoints of the displacement.
  - The work done by conservative forces is reversible because they convert between forms of mechanical energy.

- **Nonconservative forces** do work dependent on the path.
  - The work done by nonconservative forces is irreversible because they convert forms of mechanical energy to other forms of energy.

The Law of Conservation of Energy

- Thus, as long as the net work done by nonconservative forces in a system is zero, the change in total mechanical energy is zero.
- As a result, we can equate the total mechanical energy at different points in a system.
- Thus, work done by nonconservative forces should be subtracted.

Definition: Law of Conservation of Energy

Given kinetic energy K, potential energy U, and work from nonconservative forces W_{nc}

$$\Delta K + \Delta U - W_{\text{nc}} = 0$$

- For conservation of energy problems, the following form is often used:

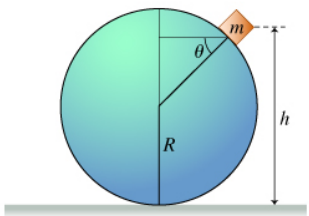
$$K_1 + U_1 + W_{\text{nc}} = K_2 + U_2$$

- Nonconservative forces may also be represented using **internal energy** U_{int} such that the internal energy changes when external forces do work on the system.

$$\Delta U_{\text{int}} = -W_{\text{nc}}$$

Example: Block on a Globe

A small block of mass m sits on top of a globe with radius R. The small block begins sliding from the top of the globe to with negligible speed. The block leaves the surface of the globe when it reaches a height h_{crit} above the ground. Derive a formula for v_{crit}, the speed of the block the moment it falls off the globe, in terms of R, h_{crit}, and g.



The first step is to identify two points in the system that would include the desired terms. One of them should be when the block is falling off because it is at height h_{crit}. The other point should be at the top of the globe because the height above the ground is just the diameter of the globe.

$$\begin{aligned}
\frac{1}{2}mv_{\text{crit}}^2 + mgh_{\text{crit}} &= mg(2R) \\
\frac{1}{2}mv_{\text{crit}}^2 &= mg(2R) - mgh_{\text{crit}} \\
v_{\text{crit}}^2 &= 4gR - 2gh_{\text{crit}} \\
v_{\text{crit}} &= \sqrt{4gR - 2gh_{\text{crit}}}
\end{aligned}$$

7.4 Force and Potential Energy

- Recall that the work done by a conservative force equals the negative change in potential energy.

$$W = -\Delta U$$

- Thus, given force $\vec{F}$ and potential energy U:

$$\vec{F} = -\vec{\nabla} U = -\frac{\partial U}{\partial x}\hat{\imath} - \frac{\partial U}{\partial y}\hat{\jmath} - \frac{\partial U}{\partial z}\hat{k}$$

The "∂" symbol refers to taking a derivative with respect to one variable, holding everything else constant, known as a *partial derivative*. Essentially, the gradient vector is a vector that takes the partial derivative of every component.